EventCode: Event-Based Barcode Identification and Paired Evaluation against RGB Cameras

Muhammad Aitsam[1][0000-0002-4157-7282], Syed Saad Hasan[2][0000-0001-8832-9027], and Alessandro Di Nuovo[1]

[1] Center of Excellence in AI and Robotics (CEAIR), Sheffield Hallam University, City Campus, S11WB, Sheffield, UK

[2] Department of Computer, Control, and Management Engineering (DIAG), Sapienza University of Rome, Via Ariosto 25, 00185 Rome, Italy

Barcodes identify products and support automated tracking in manufacturing and logistics. Reading them with conventional RGB cameras is well researched, with established methods for locating bars and decoding their widths and spacing [1]. However, rapid movement can prevent a camera from capturing these patterns clearly. A barcode moves during exposure, causing neighbouring bars and spaces to overlap in the recorded image. The reader may then misinterpret their widths or fail to decode the label. Shorter exposures reduce blur but collect less light, making reliable reading more dependent on illumination. Higher frame rates provide more opportunities to capture a readable image, but do not themselves eliminate blur within each exposure. Consequently, reliable identification requires suitable acquisition settings as well as an effective decoder. Event cameras offer an alternative. Instead of capturing complete images at fixed intervals, individual pixels asynchronously report brightness changes, recording their location, timestamp and whether brightness increased or decreased [2,3]. Their microsecond-scale temporal resolution can preserve transitions between moving bars and spaces that become blurred in RGB images. However, conventional barcode decoders cannot directly process this stream of brightness changes.

We present EventCode, an event-native method for identifying moving Code 128-B barcodes without reconstructing intensity video. To the best of our knowledge, this is the first replayable paired evaluation comparing event-native barcode identification with RGB decoding on the same moving barcode passages across multiple velocities and RGB acquisition settings. EventCode groups events into 5-ms windows and separately counts brightness increases and decreases along five lines across the barcode. The resulting spatial patterns are compared with a fixed catalogue, allowing for differences in position, polarity alignment and apparent barcode size. Evidence is combined across twenty observation windows with high event counts, indicating substantial recorded brightness changes. Identification requires repeated agreement and a sufficiently stronger match than the next-best candidate. The catalogue contains ten valid Code 128-B patterns, each incorporating a check character that helps detect decoding errors [4]. Four labels appear in the recordings, while six additional labels act as distractors, providing alternative candidates that the method must distinguish from the displayed label. The expected identity is withheld until scoring. EventCode therefore identifies registered labels rather than decoding arbitrary barcode contents.

A CenturyArks SilkyEvCam Gen3.1 and an Intel RealSense D435i observed seven-character barcodes moving across the same 59.98-Hz LCD [5,6]. Eight velocities be-

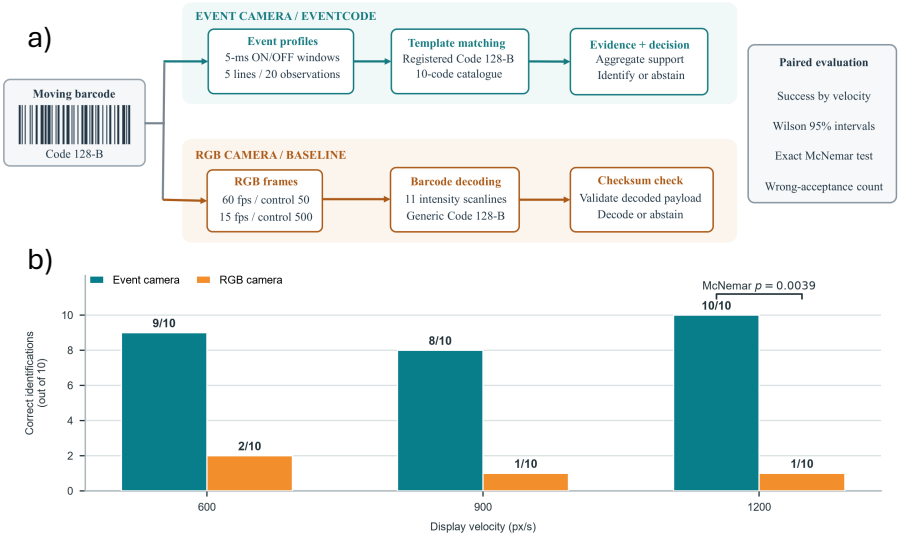


Fig. 1. (a) Paired acquisition and processing pipelines for EventCode and RGB barcode reading. (b) Correct identifications over ten trials per velocity under long-exposure RGB acquisition (15 fps, exposure-control value 500). The paired difference at 1200 pixels/s is significant under an exact McNemar test (p = 0.0039).

tween 600 and 3000 display pixels/s were tested, with ten randomised repetitions under each of two RGB configurations. These combined 60 fps with exposure-control value 50, and 15 fps with value 500. The resulting 160 paired passages were supplemented by static, blank and black controls. The RGB decoder sampled eleven lines across each barcode image and required a valid checksum before accepting a decoded payload. Raw events and lossless frames allow both fixed pipelines to be replayed. Evaluation distinguished correct identifications, wrong labels and abstentions, using Wilson confidence intervals and paired exact McNemar tests [7,8]. Under the shorter-exposure, 60-fps configuration, RGB correctly decoded all 80 moving passages. Under the longer-exposure, 15-fps configuration, EventCode achieved 9/10, 8/10 and 10/10 correct identifications at 600, 900 and 1200 pixels/s, respectively, compared with RGB results of 2/10, 1/10 and 1/10. The paired difference at 1200 pixels/s was significant (p = 0.0039). Static barcodes remained RGB-readable in both configurations, supporting a motion-related explanation for the failures. Performance did not decrease consistently with speed. Discrete LCD updates changed how far the barcode advanced between displayed images, complicating interpretation of the speed-dependent results. The findings therefore compare the tested systems and configurations rather than isolating sensor performance. This work demonstrates direct identification of moving barcodes from events, with an advantage under the tested longer-exposure, lower-frame-rate configuration. Shorter-exposure, higher-frame-rate RGB remained more reliable overall. Future work will examine physically moving printed barcodes with hardware-synchronised acquisition and extend the method towards unrestricted decoding.

References

1. Santosa, F., Goh, M.: Bar code decoding in a camera-based scanner: Analysis and algorithm. SIAM J. Imaging Sci. **15**(3), 1017–1040 (2022). https://doi.org/10.1137/21M1449658
2. Lichtsteiner, P., Posch, C., Delbruck, T.: A 128×128 120 dB 15 μs latency asynchronous temporal contrast vision sensor. IEEE J. Solid-State Circuits 43 (2), 566–576 (2008). https://doi.org/10.1109/JSSC.2007.914337
3. Gallego, G., Delbruck, T., Orchard, G., et al.: Event-based vision: A survey. IEEE Trans. Pattern Anal. Mach. Intell. 44 (1), 154–180 (2022). https://doi.org/10.1109/TPAMI.2020.3008413
4. ISO/IEC: Information technology – Automatic identification and data capture techniques – Code 128 bar code symbology specification. ISO/IEC 15417:2007, 2nd edn. (2007), confirmed 2021
5. CenturyArks Co., Ltd.: SilkyEvCam (VGA) product specifications. `https://centuryarks.com/en/silkyevcam-vga/`, last accessed 2026/09/29
6. RealSense, Inc.: RealSense D400 Series Product Family Datasheet, rev. 021 (2025). `https://dev.realsenseai.com/docs/realsense-stereo-series-product-family-datasheet/`, last accessed 2026/09/29
7. Wilson, E.B.: Probable inference, the law of succession, and statistical inference. J. Am. Stat. Assoc. **22**(158), 209–212 (1927). https://doi.org/10.1080/01621459.1927.10502953
8. McNemar, Q.: Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika **12**(2), 153–157 (1947). https://doi.org/10.1007/BF02295996

Acknowledgement: This work was supported by the UK Engineering and Physical Sciences Research Council (Grant Number EP/X018733/1 for the project ALDENS) and Innovate UK (Grant Number 10089807 for the Horizon Europe project PRIMI, Grant Agreement No. 101120727)